

## Subject Description Form

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| <b>Subject Code</b>                                   | EIE3127                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Subject Title</b>                                  | Artificial Intelligence Enabled Internet of Things                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Credit Value</b>                                   | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Level</b>                                          | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Pre-requisite/<br/>Co-requisite/<br/>Exclusion</b> | Pre-requisite:<br>EIE2112 Foundation Techniques in Artificial Intelligence <b><u>AND</u></b><br>EIE2113 Introduction of IoT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Objectives</b>                                     | <ol style="list-style-type: none"> <li>1. To introduce major application scenarios of artificial intelligence (AI) in Internet of things (IoT) and their societal impacts.</li> <li>2. To introduce AI techniques for IoT applications.</li> <li>3. To introduce communication, networking, and computing technologies in AI enabled IoT.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Intended Subject Learning Outcomes</b>             | <p><b>Upon completion of the subject, students will be able to:</b></p> <p><u>Category A: Professional/academic knowledge and skills</u></p> <ol style="list-style-type: none"> <li>1. Understand key features of IoT systems and design principles.</li> <li>2. Understand AI techniques, cloud/edge computing platforms, and wireless communication and networking techniques for AI enabled IoT.</li> <li>3. Understand key application scenarios of AI in IoT and their social impacts.</li> </ol> <p><u>Category B: Attributes for all-roundedness</u></p> <ol style="list-style-type: none"> <li>4. Think critically and creatively.</li> <li>5. Assimilate new technological development in related field.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Subject Synopsis/<br/>Indicative Syllabus</b>      | <ol style="list-style-type: none"> <li>1. <u>IoT Basics</u>: Key features of IoT; Applications of IoT; Market and ecosystem of IoT.</li> <li>2. <u>Communication and Networking for IoT</u>: Wireless communications and networking for IoT; Communication standards and protocols for IoT, IoT for massive machine type communications</li> <li>3. <u>Machine Learning for IoT</u>: Introduction of basic machine learning techniques for basic AI enabled IoT.</li> <li>4. <u>IoT Devices</u>: Onboard processors; Onboard sensors; Communication modules; AI accelerators.</li> <li>5. <u>Cloud/edge Computing for AI enabled IoT</u>: Cloud computing platforms; edge computing platforms, federated learning for AI enabled IoT</li> <li>6. <u>Security and privacy for AI enabled IoT</u>: AI enabled IoT attacks and defense, AI enabled IoT platform security, AI voice interface security, AI image interface security.</li> <li>7. <u>AI enabled IoT application scenarios</u>: Smart City; Industrial automation; Internet of vehicles. For example: <ul style="list-style-type: none"> <li>- Water quality monitoring</li> <li>- Human activity monitoring</li> <li>- Communication channel estimation in Internet of vehicles</li> </ul> </li> </ol> |

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| <b>Teaching/Learning Methodology</b>                                           | The basic features of AI enabled IoT will be described and explained in lectures. Supporting techniques, including wireless communication and networking, cloud/edge computing, as well as machine learning techniques, will be presented in lectures and tutorials. The application scenarios of AI enabled IoT will be introduced in lectures. Tutorial and lab sessions will be conducted to deliver hands-on skills on AI enabled IoT applications. The assignments and lab exercises will help students review the knowledge taught in class.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                |                                    |   |                  |   |   |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |               |     |   |   |   |   |  |              |     |   |   |   |  |  |               |     |  |  |   |   |   |                 |     |  |  |   |   |   |       |      |  |  |  |  |  |
|                                                                                | Teaching/Learning Methodology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                | Intended Subject Learning Outcomes |   |                  |   |   |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |               |     |   |   |   |   |  |              |     |   |   |   |  |  |               |     |  |  |   |   |   |                 |     |  |  |   |   |   |       |      |  |  |  |  |  |
|                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                | 1                                  | 2 | 3                | 4 | 5 |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |               |     |   |   |   |   |  |              |     |   |   |   |  |  |               |     |  |  |   |   |   |                 |     |  |  |   |   |   |       |      |  |  |  |  |  |
|                                                                                | Lecture/ Tutorial/Test/Quiz                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                | ✓                                  | ✓ | ✓                |   |   |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |               |     |   |   |   |   |  |              |     |   |   |   |  |  |               |     |  |  |   |   |   |                 |     |  |  |   |   |   |       |      |  |  |  |  |  |
|                                                                                | Laboratory/ Mini-project                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                |                                    |   | ✓                | ✓ | ✓ |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |               |     |   |   |   |   |  |              |     |   |   |   |  |  |               |     |  |  |   |   |   |                 |     |  |  |   |   |   |       |      |  |  |  |  |  |
|                                                                                | Assignment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                | ✓                                  | ✓ | ✓                | ✓ |   |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |               |     |   |   |   |   |  |              |     |   |   |   |  |  |               |     |  |  |   |   |   |                 |     |  |  |   |   |   |       |      |  |  |  |  |  |
| <b>Assessment Methods in Alignment with Intended Subject Learning Outcomes</b> | <table><tr><td rowspan="2">Specific assessment methods/tasks</td><td rowspan="2">% weighting</td><td colspan="5">Intended subject learning outcomes to be assessed (Please tick as appropriate)</td></tr><tr><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>1. Assignment</td><td>15%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr><tr><td>2. Test/Quiz</td><td>30%</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td></tr><tr><td>3. Laboratory</td><td>20%</td><td></td><td></td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>4. Mini-project</td><td>35%</td><td></td><td></td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Total</td><td>100%</td><td colspan="5"></td></tr></table> <p><b>Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:</b></p> <p>Assignment, test, and quiz let students review the taught materials, do further reading for deeper learning and apply the learnt materials to solving problems for AI enabled IoT.</p> <p>Laboratory and mini-project require students to do further reading, search for information, and develop AI enabled IoT applications.</p> |                                                                                |                                    |   |                  |   |   | Specific assessment methods/tasks | % weighting | Intended subject learning outcomes to be assessed (Please tick as appropriate) |  |  |  |  | 1 | 2 | 3 | 4 | 5 | 1. Assignment | 15% | ✓ | ✓ | ✓ | ✓ |  | 2. Test/Quiz | 30% | ✓ | ✓ | ✓ |  |  | 3. Laboratory | 20% |  |  | ✓ | ✓ | ✓ | 4. Mini-project | 35% |  |  | ✓ | ✓ | ✓ | Total | 100% |  |  |  |  |  |
| Specific assessment methods/tasks                                              | % weighting                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Intended subject learning outcomes to be assessed (Please tick as appropriate) |                                    |   |                  |   |   |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |               |     |   |   |   |   |  |              |     |   |   |   |  |  |               |     |  |  |   |   |   |                 |     |  |  |   |   |   |       |      |  |  |  |  |  |
|                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1                                                                              | 2                                  | 3 | 4                | 5 |   |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |               |     |   |   |   |   |  |              |     |   |   |   |  |  |               |     |  |  |   |   |   |                 |     |  |  |   |   |   |       |      |  |  |  |  |  |
| 1. Assignment                                                                  | 15%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | ✓                                                                              | ✓                                  | ✓ | ✓                |   |   |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |               |     |   |   |   |   |  |              |     |   |   |   |  |  |               |     |  |  |   |   |   |                 |     |  |  |   |   |   |       |      |  |  |  |  |  |
| 2. Test/Quiz                                                                   | 30%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | ✓                                                                              | ✓                                  | ✓ |                  |   |   |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |               |     |   |   |   |   |  |              |     |   |   |   |  |  |               |     |  |  |   |   |   |                 |     |  |  |   |   |   |       |      |  |  |  |  |  |
| 3. Laboratory                                                                  | 20%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                |                                    | ✓ | ✓                | ✓ |   |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |               |     |   |   |   |   |  |              |     |   |   |   |  |  |               |     |  |  |   |   |   |                 |     |  |  |   |   |   |       |      |  |  |  |  |  |
| 4. Mini-project                                                                | 35%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                |                                    | ✓ | ✓                | ✓ |   |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |               |     |   |   |   |   |  |              |     |   |   |   |  |  |               |     |  |  |   |   |   |                 |     |  |  |   |   |   |       |      |  |  |  |  |  |
| Total                                                                          | 100%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                |                                    |   |                  |   |   |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |               |     |   |   |   |   |  |              |     |   |   |   |  |  |               |     |  |  |   |   |   |                 |     |  |  |   |   |   |       |      |  |  |  |  |  |
| <b>Student Study Effort Expected</b>                                           | <b>Class contact (time-tabled):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                |                                    |   |                  |   |   |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |               |     |   |   |   |   |  |              |     |   |   |   |  |  |               |     |  |  |   |   |   |                 |     |  |  |   |   |   |       |      |  |  |  |  |  |
|                                                                                | • Lecture/Tutorial                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                |                                    |   | 18 Hours         |   |   |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |               |     |   |   |   |   |  |              |     |   |   |   |  |  |               |     |  |  |   |   |   |                 |     |  |  |   |   |   |       |      |  |  |  |  |  |
|                                                                                | • Laboratory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                |                                    |   | 12 Hours         |   |   |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |               |     |   |   |   |   |  |              |     |   |   |   |  |  |               |     |  |  |   |   |   |                 |     |  |  |   |   |   |       |      |  |  |  |  |  |
|                                                                                | • Mini-project development                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                |                                    |   | 9 Hours          |   |   |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |               |     |   |   |   |   |  |              |     |   |   |   |  |  |               |     |  |  |   |   |   |                 |     |  |  |   |   |   |       |      |  |  |  |  |  |
|                                                                                | <b>Other student study effort:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                |                                    |   |                  |   |   |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |               |     |   |   |   |   |  |              |     |   |   |   |  |  |               |     |  |  |   |   |   |                 |     |  |  |   |   |   |       |      |  |  |  |  |  |
|                                                                                | • Lecture: preview/review of notes; assignment; preparation for test/quiz                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                |                                    |   | 30 Hours         |   |   |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |               |     |   |   |   |   |  |              |     |   |   |   |  |  |               |     |  |  |   |   |   |                 |     |  |  |   |   |   |       |      |  |  |  |  |  |
|                                                                                | • Tutorial/Laboratory/Mini-project: preview of materials, revision and/or reports writing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                |                                    |   | 36 Hours         |   |   |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |               |     |   |   |   |   |  |              |     |   |   |   |  |  |               |     |  |  |   |   |   |                 |     |  |  |   |   |   |       |      |  |  |  |  |  |
|                                                                                | <b>Total student study effort:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                |                                    |   | <b>105 Hours</b> |   |   |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |               |     |   |   |   |   |  |              |     |   |   |   |  |  |               |     |  |  |   |   |   |                 |     |  |  |   |   |   |       |      |  |  |  |  |  |

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|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Reading List and References</b> | <p><b>Textbook:</b></p> <ol style="list-style-type: none"> <li>1. Jan Holler, Vlasios Tsiatsis, Catherine Mulligan, Stefan Avesand, Stamatis Karnouskos and David Boyle, <i>From Machine-To-Machine to the Internet of Things: Introduction to a New Age of Intelligence</i>, Academic Press, 2014.</li> <li>2. Gurjit Kaur, Pradeep Tomar, and Marcus Tanque, <i>Artificial Intelligence to Solve Pervasive Internet of Things Issues</i>, Academic Press, 2020.</li> <li>3. Yaser S. Abu-Mostafa, Malik Magdon-Ismael, and Hsuan-Tien Lin, <i>Learning from Data</i>, AMLBook, 2017.</li> </ol> <p><b>Reference Materials:</b></p> <ol style="list-style-type: none"> <li>1. J. Gubbi, R. Buyya, S. Marusic, and M. Palaniswami, "Internet of Things (IoT): A vision, architectural elements, and future directions," <i>Future Gener. Comput. Syst.</i>, vol. 29, no. 7, pp. 1645–1660, Sep. 2013.</li> <li>2. A. A. Fuqaha, M. Guizani, M. Mohammadi, M. Aledhari, and M. Ayyash, "Internet of Things: A survey on enabling technologies, protocols, and applications," <i>IEEE Commun. Surveys Tuts.</i>, vol. 17, no. 4, pp. 2347–2376, 4th Quart., 2015.</li> </ol> |
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